

Muneeb Abbasi

Principal AI Engineer — LLMs, RAG & agentic systems · 15+ years · ex-Microsoft

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Summary

- Principal Software Engineer with 15+ years delivering enterprise-scale software, AI platforms, cloud-native systems, and workflow automation, leading distributed in-house and offshore engineering teams
- Proven ability to translate complex business challenges into scalable technical solutions across AI, cloud, distributed systems, and enterprise integrations
- Hands-on expertise in LLMs, RAG, AI agents, agent evaluation and observability, AWS (Bedrock, SageMaker, Lambda, EKS), Azure, Java / Spring Boot, C# / .NET, Python / FastAPI, React, microservices, and data platforms
- Experienced in customer engagement, technical discovery, solution architecture, and end-to-end delivery from requirements through production adoption
- Recognized for combining hands-on engineering, technical leadership, and measurable business outcomes

Technical expertise

AI & applied intelligence	Generative AI, LLMs, Retrieval-Augmented Generation (RAG), AI agents, agentic workflows, Model Context Protocol (MCP), semantic & vector search, prompt engineering, Amazon Bedrock (Claude, Titan, Llama), AWS Knowledge Bases (RAG), Titan embeddings, Amazon SageMaker, Textract, Comprehend, AI guardrails (PII, prompt-injection, tool allow-lists, action approval), AI evaluation & observability (golden datasets, groundedness/hallucination analysis, shadow & canary testing, tracing, prompt/model versioning), OpenAI APIs, Azure OpenAI, LangGraph, scikit-learn
Cloud & infrastructure	AWS (Lambda, Step Functions, EventBridge, SNS/SQS, EKS/ECS, S3, API Gateway, Cognito, CloudFront, CloudWatch, X-Ray, CloudTrail), Microsoft Azure, Azure Functions, Docker, Kubernetes, event-driven architecture, blue/green & canary deployments, AWS CDK (IaC), CodePipeline, Multi-AZ & cross-region DR (pilot light), IAM Identity Center, KMS, WAF
Software engineering	Java, Spring Boot, Python, FastAPI, C#, .NET Core, TypeScript, JavaScript, React Native, SQL
Data & storage	Amazon Aurora PostgreSQL (pgvector), DynamoDB, Amazon Redshift, OpenSearch, ElastiCache (Redis), SQL Server, Azure SQL, MongoDB, vector databases, AWS Glue, Amazon Athena, Amazon QuickSight, data pipelines
Frontend	React, TypeScript, React Native, Angular, HTML5, enterprise dashboards, data visualization
DevOps & tooling	Git, GitHub Actions, Azure DevOps, Jira, Visual Studio, VS Code, IntelliJ IDEA
AI development tools	Claude Code, GitHub Copilot, AI-assisted development (design, code generation, testing), rigorous review of AI-generated output

Professional experience

Principal AI Engineer — Global E-commerce & Retail

Bellevue, WA · Feb 2025 – Present

AI-Powered Inventory & Operations Platform — architect and hands-on AI lead of an enterprise AI platform on AWS using LLMs, RAG, AI agents, semantic search, and machine learning to modernize inventory, supply chain, procurement, logistics, sales, and customer operations through a scalable, event-driven, cloud-native architecture.

- Own the platform architecture end-to-end, leading an 18-engineer program (3 in-house + 15 offshore across domain, AI-platform, frontend/mobile, and DevOps/QA pods) from discovery through production
- Designed and delivered the AI platform leveraging Amazon Bedrock (Claude, Titan, Llama), RAG with AWS Knowledge Bases over an OpenSearch knowledge store and Aurora pgvector catalog embeddings, Titan

embeddings and reranking, and Textract/Comprehend document intelligence, reducing manual reconciliation by 90%

- Built seven domain AI agents (quote, procurement, inventory & warehouse, delivery-exception, warranty, finance, executive) coordinated through an AI gateway -> orchestrator -> model router -> tool registry chain with tool/function calling and human-in-the-loop approval, protected by a defense-in-depth guardrails layer (PII protection, prompt-injection protection, per-agent tool allow-lists, action approval, audit logging)
- Built a four-layer evaluation harness (retrieval, generation, agent/tool behavior, business outcome) with a ~1,200-case golden dataset, regression gates, shadow and canary release, end-to-end tracing, and prompt/model/index versioning
- Architected event-driven microservices across 18+ business domains using Java / Spring Boot (business services) and Python/FastAPI (AI services) on EventBridge, SNS/SQS, Step Functions, and Lambda, containerized on EKS/ECS, with transactional outbox, idempotent consumers, circuit breakers, and schema-registry patterns
- Delivered predictive ML on Amazon SageMaker (demand forecasting, supplier-delay risk, delivery ETA, anomaly detection) and the data layer across Aurora PostgreSQL, DynamoDB, Redshift, S3, OpenSearch, and ElastiCache, with CloudWatch/X-Ray/CloudTrail observability, blue/green and canary deploys, 99.9%/99.5% SLOs, and cross-region pilot-light DR

Senior Software Engineer — Microsoft

Redmond, WA · Oct 2014 – Jan 2025

Partnered with engineering organizations and infrastructure stakeholders to solve enterprise-scale operational challenges — leading technical discovery, proposing solution architectures, and delivering cloud-native automation and AI-powered platforms from requirements through production as a hands-on engineer.

Network Device Automation System

- Designed and developed scalable cloud-native microservices using .NET Core, Azure Functions, and event-driven architectures to orchestrate device scheduling, deployment, validation, and monitoring across global infrastructure
- Built monitoring, validation, and workflow orchestration that improved deployment reliability and operational visibility
- Reduced manual network device scheduling effort by 60% through intelligent workflow automation, achieving 99.9% platform availability

NIC — Network Infrastructure Copilot & Enterprise AI Knowledge Platform

- Developed enterprise AI copilots using LLMs, RAG, semantic search, and AI orchestration enabling natural-language interaction with telemetry, KPIs, incidents, documentation, and operational systems
- Led technical discovery and proposed enterprise AI architectures, integrating multiple enterprise repositories and operational systems securely
- Increased engineering productivity by 35% through AI-powered natural-language access to enterprise knowledge
- Reduced engineering research and incident-investigation time by 60%, accelerating resolution across global infrastructure

Projects

- Career Vault — agentic job-search platform (github.com/amuneeb/career-vault-demo): GitHub Actions pipeline, fit-scoring, ATS-email status tracking, Cloudflare-edge React dashboard, MCP-integrated Claude-agent application workflow — \$0 infrastructure

Education

- Master of Business Information Systems (MBS) — University College Cork · 2006–2007
- Bachelor of Engineering in Information Technology (B.Eng.) — Hamdard University · 1999–2003

Awards

- Sogeti Star Award — Sogeti (2012) · Innovative Employee Award — STRATEGEZ (2005)